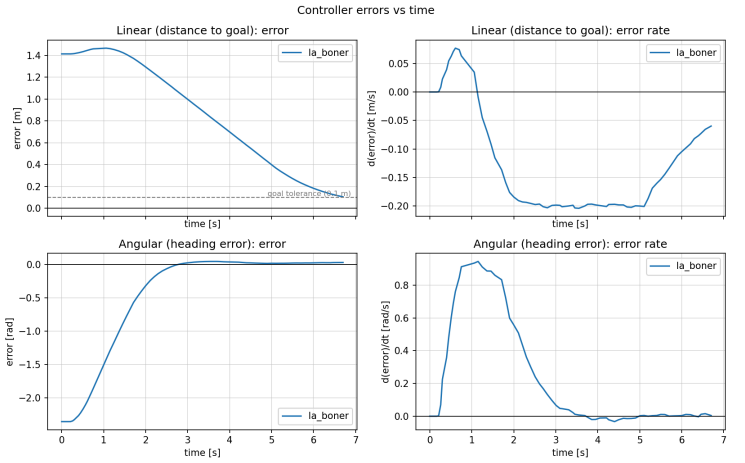
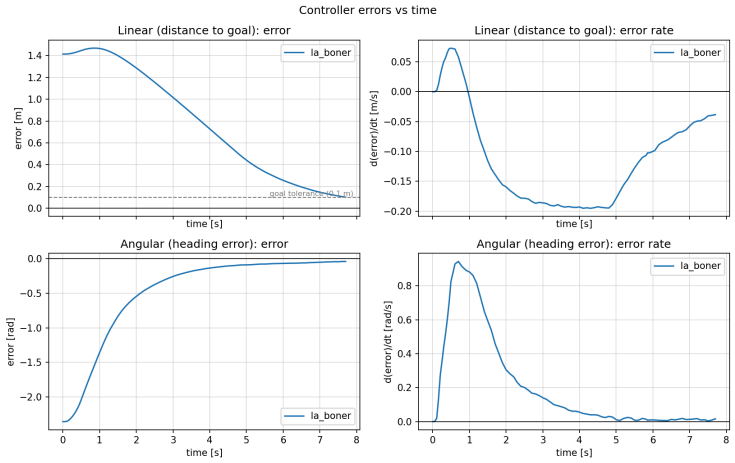
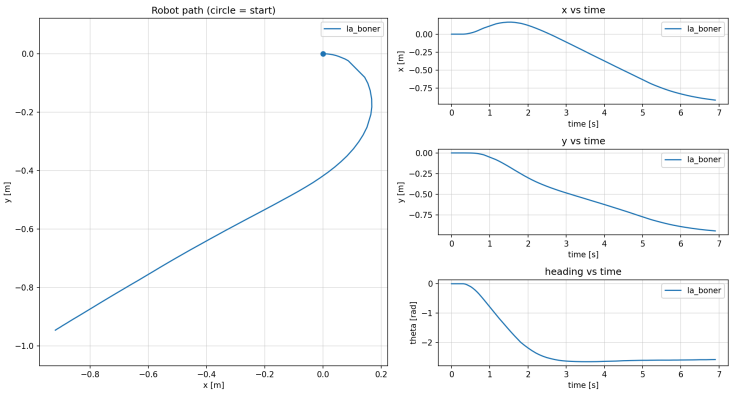
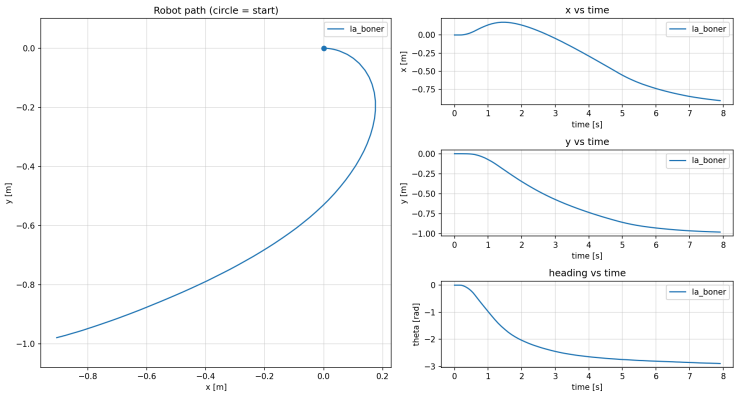
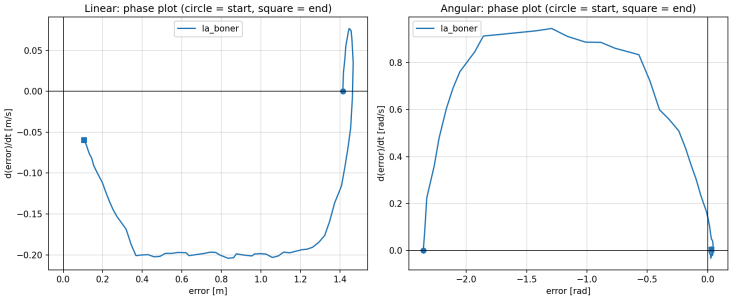
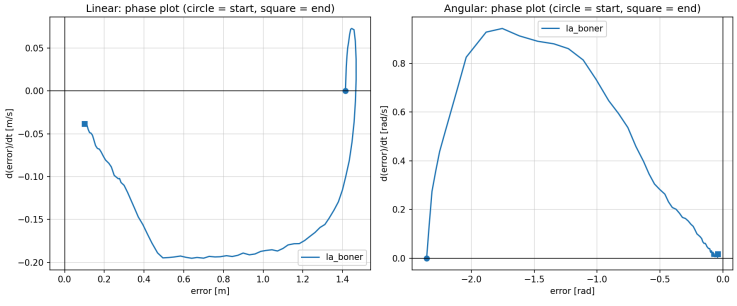
MTE 544 Lab 1

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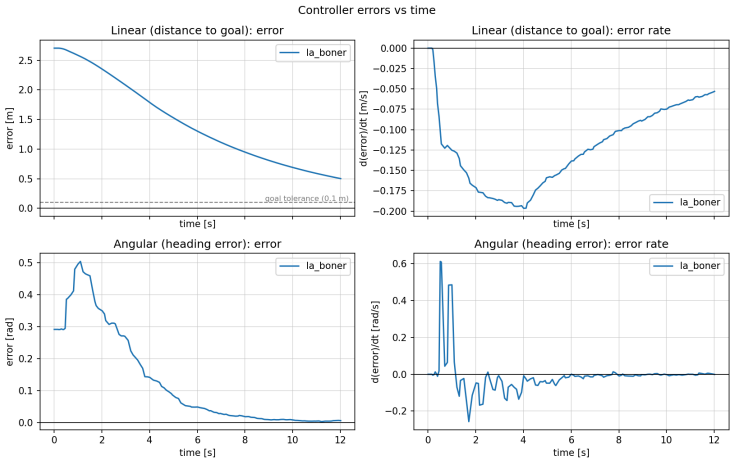
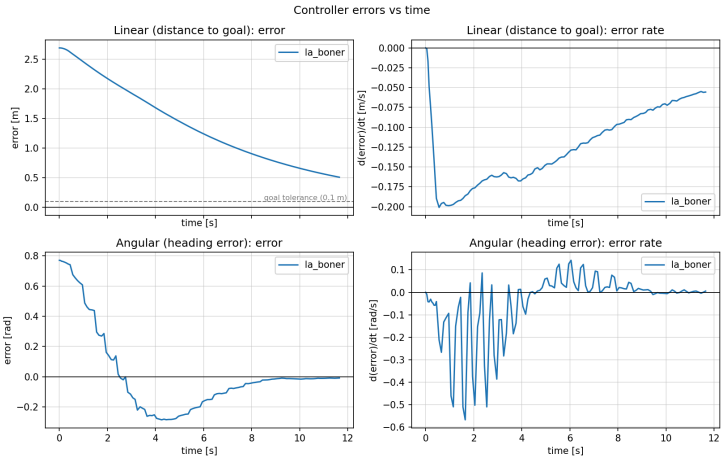
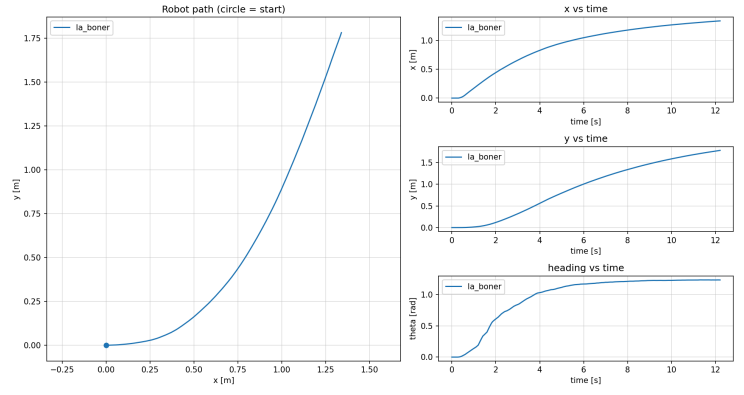
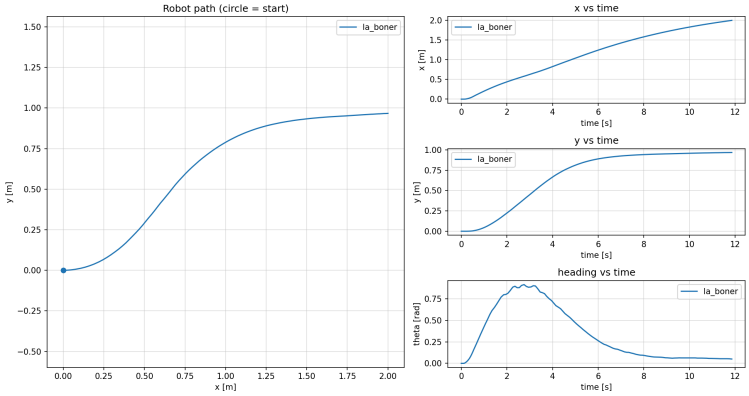
Robot Number: 6
Station Number: MME PC164

Section 1

The plots below were generated with a tuned P and PID controller on the [-1, -1] point goal.

P Controller	PID Controller
Error and Error Rate vs. Time	
	
X, Y, Theta vs Time, along with XY space graph	
	
Linear and Angular Error vs. Rate of Change of Error	
	

Based on the plots, the PID controller exhibits a better overall response than the P controller. This is largely based on the improvement in angular error performance, as the P controller's angular error displayed overshoot and oscillations as it settled towards its steady-state value. The PID controller, however, more closely resembled a critically damped system, and while it may be slightly overdamped, the lack of overshoot and smoother response makes it more desirable. Based on this observation, the PID controller was used for the parabola and sigmoid trajectories. These trajectories and their corresponding error plots can be seen below. Both trajectories were followed with the error plots exhibiting that same damped behaviour as seen with the point goal.

Parabola Trajectory using PID Controller	**Sigmoid Trajectory using PID Controller**
Error and Error Rate vs. Time	
	
X, Y, Theta vs Time, along with XY space graph	
	

Section 2

Values calculated using angular position and timestamp data.

Controller	Metric	P	PID
Agility (Rise time & 5% settling time)	Heading rise time (90% to 10% of initial error)	1.65s	2.75s
	Heading settling time (±5% band)	2.40 s	4.30 s
	Linear rise time (90% to 10% of initial error)	4.29 s	5.00 s
	Time to reach the 0.1m stopping distance	6.70 s	7.70 s
Accuracy (Steady state error)	Mean absolute heading error, final 1s before stopping	0.027 rad (1.53°)	0.045 rad (2.58°)
	Final distance to goal (set by the 0.1m stopping rule)	0.097 m	0.096 m
Overshoot (OS%)	Heading overshoot	1.95% (0.046 rad)	0%
	Linear overshoot	Not measurable	Not measurable

To measure agility, the rise time and 5% settling time from both the linear and angular data sets were used for the point controller. The numerical values for both rise time and 5% settling time show that the P controller performs slightly better in terms of Agility.

To measure accuracy, the steady-state error was used from both the angular and linear data sets. For the linear portion of the robot motion, a deadband of 0.1m was created to limit the oscillation as the robot tries to reach the desired position. This means the steady-state error for the linear portion will always be around 0.1m and is not very useful in comparing the P and PID controllers. The angular data set is a better representation of the differences between controllers and shows that both controllers perform similarly in this category. The difference in performance of P and PID controllers could be due to run-to-run error; therefore, there isn't a clear winner in the accuracy category.

To measure overshoot, the angular data set was used because the linear portion of the robot's movement was always stopped around 0.1m before it reached the target due to the deadband. The PID controller performs better than the P controller in this category.